

Big Data Tools: KNIME, Spark, and Databricks

Lecture 6: Scalable Data Science with Free Tools (MSc Data Science)

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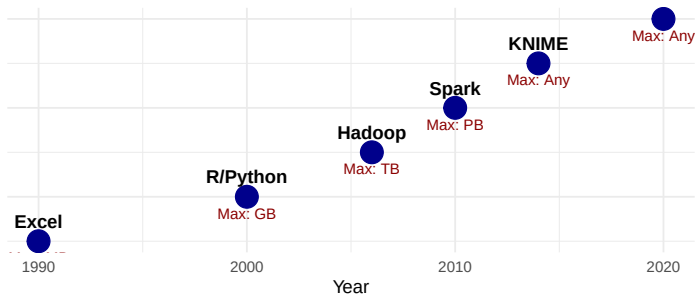
Section 1

Part I: The Big Data Challenge and KNIME Workflows

Slide 1: The Big Data Revolution

Evolution of Data Science Tools:

Evolution of Data Science Tools
From desktop to distributed computing



Slide 2: Why Traditional Tools Break Down

The Scaling Problem:

Data Size	Tool	Processing Time	Memory
1 GB	R (laptop)	Minutes	8 GB RAM
10 GB	R (laptop)	Hours	32 GB RAM
100 GB	R (laptop)	Crash!	Not enough
100 GB	Spark (cluster)	Minutes	Distributed
1 TB	Spark (cluster)	Hours	Distributed

Key Insight: When data doesn't fit in memory, you need distributed computing

Three Free Tools for Big Data:

① KNIME Analytics (Slides 1-40):

- Visual workflow designer (no-code)
- Runs on laptop
- Perfect for learning and prototyping

② Apache Spark + Databricks CE (Slides 41-100):

- Distributed computing framework
- Free cloud tier (15 GB RAM)
- Industry standard

③ Integration (Slides 101-150):

- Combining KNIME, R, and Python
- Deployment strategies

Slide 4: What is KNIME?

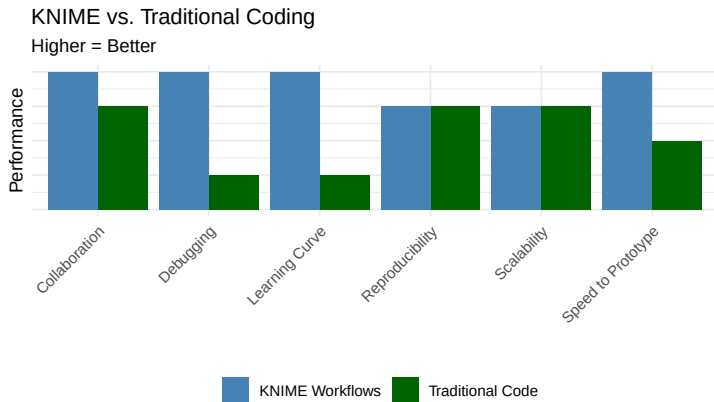
KNIME Analytics Platform:

- **K**onstanz **N**formation **M**iner
- Open-source, visual workflow tool
- Drag-and-drop interface (no coding required!)
- 2000+ pre-built nodes (operations)

Key Advantages:

- Free and open-source
- Runs on Windows, Mac, Linux
- Integrates R, Python, SQL, Spark
- Visual pipelines (easy to understand and share)
- Production-ready (can deploy workflows)

Slide 5: KNIME vs. Traditional Coding



Slide 6: Installing KNIME Analytics Platform

Installation Steps:

- 1 Visit: **<https://www.knime.com/downloads>**
- 2 Download KNIME Analytics Platform (free)
- 3 Install (no license needed)
- 4 Launch KNIME

System Requirements:

- Windows 10+, macOS 10.15+, or Linux
- 4 GB RAM minimum (8 GB recommended)
- 2 GB disk space
- Java 11+ (included in installer)

First Launch: Creates workspace folder for your workflows

Slide 7: KNIME Interface Overview

Main Components:

- ➊ **Node Repository (Left):** Library of 2000+ operations
- ➋ **Workflow Canvas (Center):** Drag nodes here
- ➌ **Workflow Coach (Right):** Suggests next steps
- ➍ **Description (Bottom):** Node documentation
- ➎ **Console (Bottom):** Execution messages

Key Terms:

- **Node:** Single operation (read file, filter, model)
- **Workflow:** Connected nodes (complete pipeline)
- **Port:** Connection point (triangle = data table)
- **Configure:** Set node parameters (double-click)
- **Execute:** Run node (right-click → Execute)

Slide 8: Your First KNIME Workflow - Hello Data

Simple 3-Node Workflow:

[Data Generator] → [Row Filter] → [Table View]

Steps:

❶ Node Repository → Manipulation → Row → Data Generator

- Drag to canvas
- Configure: 100 rows
- Execute (green light = success)

❷ Manipulation → Row → Row Filter

- Drag to canvas, connect to Data Generator
- Configure: Keep rows where Column0 > 50
- Execute

❸ Views → Table View

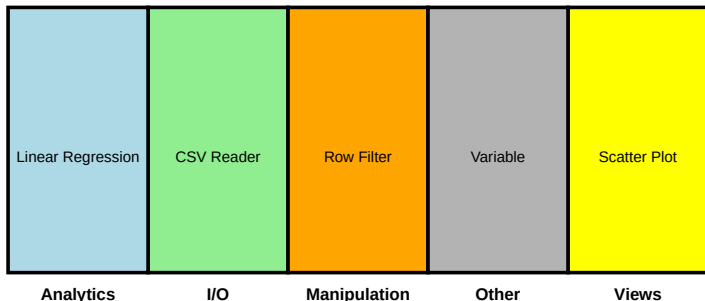
- Connect and execute
- View results

Slide 9: KNIME Node Types - Color Coding

Node Colors Indicate Function:

KNIME Node Color Coding

Colors help identify node categories



Port Shapes: Triangle = data table, Square = model, Circle = other

Slide 10: Reading Data in KNIME

Common Data Sources:

① CSV Reader (most common)

- File → CSV Reader
- Browse to file
- Auto-detects delimiter, headers

② Excel Reader

- Reads .xlsx files
- Select specific sheets

③ Database Connector

- MySQL, PostgreSQL, SQLite
- Execute SQL queries

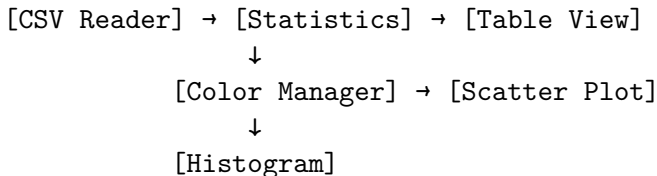
④ File Reader (generic)

- Auto-detects format

Demo: Read iris.csv dataset

Slide 11: Data Exploration Nodes

Essential Exploration Nodes:



Key Nodes:

- **Statistics:** Mean, median, min, max per column
- **Table View:** Browse data (like View() in R)
- **Histogram:** Distribution of numeric columns
- **Scatter Plot:** Relationships between variables
- **Box Plot:** Outlier detection
- **Missing Value:** Check for NAs

Slide 12: Data Cleaning Workflow

Common Cleaning Operations:

[CSV Reader]



[Column Filter] (remove unwanted columns)



[Row Filter] (remove outliers/bad data)



[Missing Value] (handle NAs)



[String Manipulation] (clean text)



[Normalizer] (scale numeric features)



[Column Rename] (standardize names)

Best Practice: Chain nodes to create reproducible cleaning pipeline

Slide 13: Feature Engineering in KNIME

Creating New Features:

1 Math Formula Node:

- Create derived columns
- Example: $\$price\$ / \$area\$ = price_per_sqm$

2 String Manipulation:

- Extract substrings
- Convert case
- Replace patterns

3 Rule Engine:

- If-then-else logic
- Example: $\$age\$ > 65 \Rightarrow \text{"Senior"}$

4 Java Snippet:

- Custom code for complex operations

Slide 14: Partitioning Data - Train/Test Split

Node: Partitioning

[CSV Reader]



[Partitioning] (80% train / 20% test)

↓ (two outputs)

[Training Data]

[Test Data]

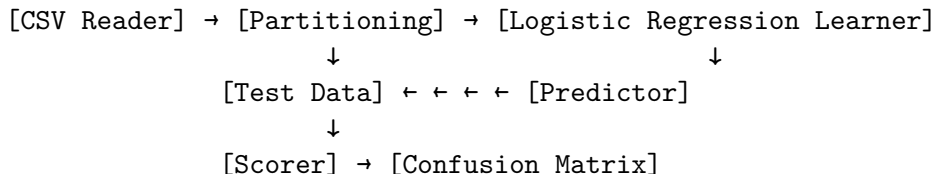
Configuration:

- **Relative:** 80% / 20%
- **Absolute:** First 1000 rows / rest
- **Stratified:** Preserve class distribution
- **Random Sampling:** With seed for reproducibility

Important: Set random seed for reproducible splits!

Slide 15: End-to-End Classification Workflow - Overview

Complete Iris Classification Pipeline:



Workflow Steps:

- 1 Load iris data
- 2 Split 80/20
- 3 Train logistic regression on training set
- 4 Predict on test set
- 5 Evaluate with confusion matrix

Slide 16: Step 1 - Load Iris Data

Node: CSV Reader

Configuration:

- Browse to `iris.csv`
- Check “Has Column Headers”
- Check “Has Row IDs”
- Click OK

Execute and View:

- Right-click → Execute
- Right-click → Output → Data Table
- Verify: 150 rows, 5 columns
- Species column should be categorical

If Species is not categorical: Add **String to Number** node

Slide 17: Step 2 - Partition Data

Node: Partitioning

Configuration:

- ① Drag **Manipulation** → **Row** → **Partitioning**
- ② Connect to CSV Reader
- ③ Configure:
 - Relative: 80% top / 20% bottom
 - Stratified sampling: Check
 - Column: Species
 - Random seed: 123

Why Stratified? Preserves class distribution in both sets

Execute: Node should show two output ports (top = train, bottom = test)

Slide 18: Step 3 - Train Logistic Regression

Node: Logistic Regression Learner

Path: Analytics → Mining → Logistic Regression Learner

Configuration:

- ❶ Connect top port of Partitioning node
- ❷ Double-click to configure:
 - Target column: Species
 - Feature columns: Select all numeric columns
 - Solver: Default (IRLS)

Execute:

- Green light = model trained successfully
- Model object stored in output port (square shape)

Slide 19: Step 4 - Make Predictions

Node: Predictor

Configuration:

- 1 Drag **Analytics** → **Mining** → **Predictor**
- 2 Connect TWO inputs:
 - Model from Logistic Regression Learner (square port)
 - Test data from Partitioning (bottom triangle port)
- 3 Configure:
 - Append columns with suffix: `_predicted`
 - Include probabilities: Check

Execute: Adds prediction columns to test data

Slide 20: Step 5 - Evaluate Model

Nodes: Scorer + Confusion Matrix

[Predictor] → [Scorer] → [Confusion Matrix]

Scorer Node:

- Analytics → Mining → Scorer
- Connect to Predictor output
- Configure:
 - First column: Species (actual)
 - Second column: Species_predicted
- Execute

Confusion Matrix:

- Views → Confusion Matrix
- Connect to Scorer
- Execute and view results

Slide 21: Understanding KNIME Output

Scorer Output Statistics:

- **Accuracy:** Overall correctness
- **Precision:** True positives / (TP + FP)
- **Recall:** True positives / (TP + FN)
- **F1-Score:** Harmonic mean of precision and recall

Confusion Matrix:

	Predicted Setosa	Predicted Versicolor	Predicted Virginica
Actual Setosa	TP	FP	FP
Actual Versicolor	FN	TP	FP
Actual Virginica	FN	FN	TP

Slide 22: Saving Your Workflow

Save Workflow:

- 1 File → Save As
- 2 Choose location
- 3 Name: Iris_Classification
- 4 Creates .knwf file

Export Workflow:

- 1 File → Export KNIME Workflow
- 2 Creates portable .knwf archive
- 3 Can share with others

Best Practice:

- Save frequently
- Use version control (Git-friendly)
- Add annotations (Edit → Workflow Annotations)

Slide 23: Adding Documentation to Workflows

Making Workflows Understandable:

1 Node Descriptions:

- Right-click node → Edit Node Description
- Explain what this node does

2 Workflow Annotations:

- Right-click canvas → Workflow Annotation
- Add text boxes explaining sections

3 Meta Nodes:

- Group related nodes
- Right-click nodes → Create Meta Node
- Collapse complex sections

Good Practice: Document as you build, not after!

Slide 24: KNIME Components - Reusable Modules

What are Components?

- Reusable sub-workflows
- Encapsulate complex logic
- Share across projects

Create Component:

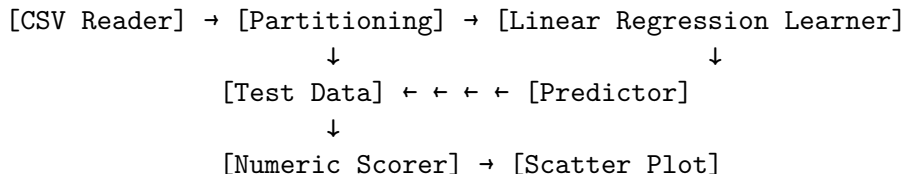
- 1 Select multiple nodes
- 2 Right-click → Create Component
- 3 Name it (e.g., “Data Cleaning”)
- 4 Configure inputs/outputs

Use Cases:

- Standard data preprocessing
- Custom visualization
- Feature engineering pipelines

Slide 25: Regression Workflow in KNIME

Predicting House Prices:



Key Differences from Classification:

- **Learner:** Linear Regression (not Logistic)
- **Scorer:** Numeric Scorer (not Classification Scorer)
- **Metrics:** RMSE, MAE, R^2 (not accuracy)

Slide 26: Regression Evaluation Nodes

Numeric Scorer Output:

[Predictor] → [Numeric Scorer]



Statistics Output:

- Mean Absolute Error (MAE)
- Root Mean Squared Error (RMSE)
- R^2 (coefficient of determination)
- Mean Signed Difference

Visualization:

- **Scatter Plot:** Actual vs. Predicted
 - Perfect predictions = diagonal line
 - Points above = over-predictions
 - Points below = under-predictions

Slide 27: Advanced Nodes - Cross-Validation

X-Partitioner + X-Aggregator:

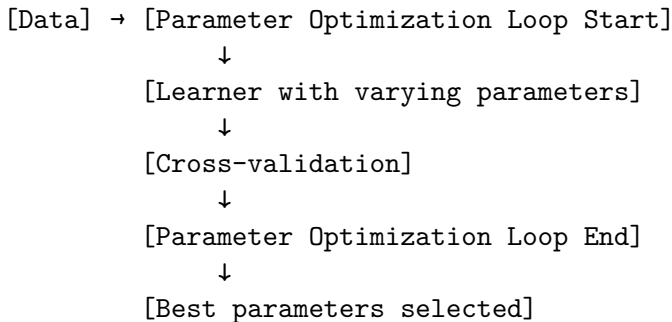
```
[Data] → [X-Partitioner (k=5)]  
      ↓ (loop)  
[Learner] → [Predictor] → [Scorer]  
      ↓ (collect results)  
[X-Aggregator]  
      ↓  
[Mean Accuracy across folds]
```

Configuration:

- X-Partitioner: Set number of folds (k=5 or k=10)
- Stratified: Preserve class distribution
- X-Aggregator: Calculates average performance

Slide 28: Parameter Optimization in KNIME

Parameter Optimization Loop:



Use Case: Find optimal hyperparameters (similar to grid search in R)

Example: Optimize regularization in Logistic Regression

Slide 29: Exporting Models from KNIME

Deployment Options:

① PMML (Predictive Model Markup Language):

- Model → PMML Writer
- Export model in standard format
- Import into R, Python, Java

② Python Node:

- Convert KNIME model to Python
- Deploy in Python environments

③ REST Service:

- KNIME Server (commercial)
- Expose workflow as API

④ Batch Scoring:

- Save workflow
- Run headless: `knime -consoleLog -nosplash -application org.knime.product.KNIME_BATCH_APPLICATION`

Slide 30: KNIME Hub - Share and Discover

KNIME Hub: hub.knime.com

Features:

- **Public Workflows:** Download examples
- **Components:** Reusable building blocks
- **Extensions:** Additional node packages
- **Community:** Ask questions, share solutions

Popular Extensions:

- **Deep Learning:** TensorFlow, Keras integration
- **Text Processing:** NLP nodes
- **Big Data:** Spark, Hadoop connectors
- **Time Series:** Specialized forecasting nodes

Next Lecture Section: Introduction to Spark and Databricks (Slides 31-60)

